

OPEN DATA CENTER FACILITY ENGINEERING

A building-only, end-to-end concept portfolio
Jonathan Oktaviano | Reference release 0.1.0 | 3 September 2026

H01 / 1.25 MW 	H02 / 1.25 MW 	H03 / 1.25 MW 	H04 / 1.25 MW 
DESIGN IT	ANNUAL PUE	CONCEPT CAPEX	IMPLEMENTED EVIDENCE
5.00 MW	1.347	USD 57.60 million	8760-hour model + six offline cases

One scenario connects requirements, power, cooling, energy, facility economics, asset identities and commissioning examples. Source code, source records and original drawings accompany this dossier.

Independent portfolio. No AWS affiliation or proprietary hyperscaler design. Synthetic inputs; concept engineering; not certified or for construction.

01 / Executive decision

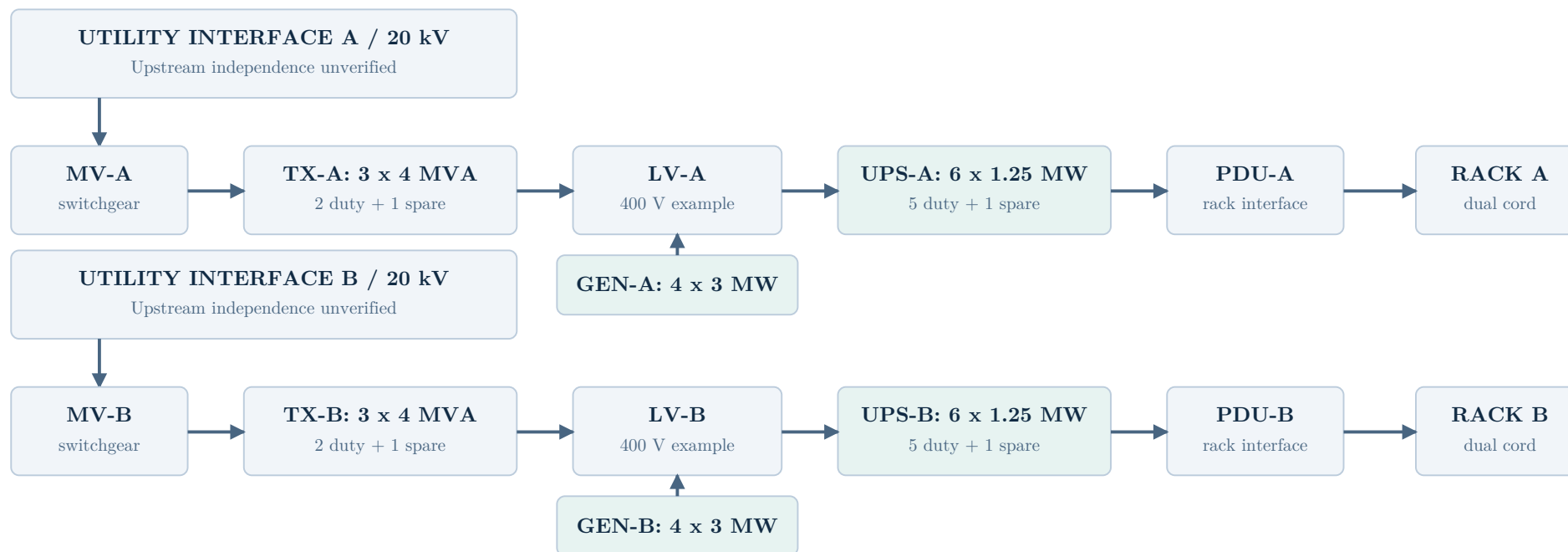
Keep the base case transparent. Its slightly negative NPV shows why engineering design and commercial assumptions must be assessed together. No real-site investment conclusion is established.

RESULT	VALUE	INTERPRETATION
Design facility input	6.814 MW	Full IT at 35 C; sizing margin applied afterward
Annual gross facility energy	44.242 GWh	Fully leased; 75% mean draw; synthetic weather
Pre-tax unlevered NPV	USD -0.329 million	15 years; 10% nominal discount
Conventional IRR	9.92%	Not a levered equity return
Simple payback	8.67 years	Undiscounted first crossing

Capital is an author-selected USD 48m base allowance plus 20% contingency. Capacity pricing is USD 180/kW-month with a 35-95% occupancy ramp. Energy reimbursement is 100%; rent and electricity recovery remain separate cash-flow lines.

Next evidence: utility connection and site hazards; equipment and contractor quotations; lease terms; construction phasing; cooling ride-through; detailed isolation and protection design.

02 / Electrical concept and capacity



Concept block SLD | breaker, protection, earthing, ties and auxiliary circuits require detailed design.

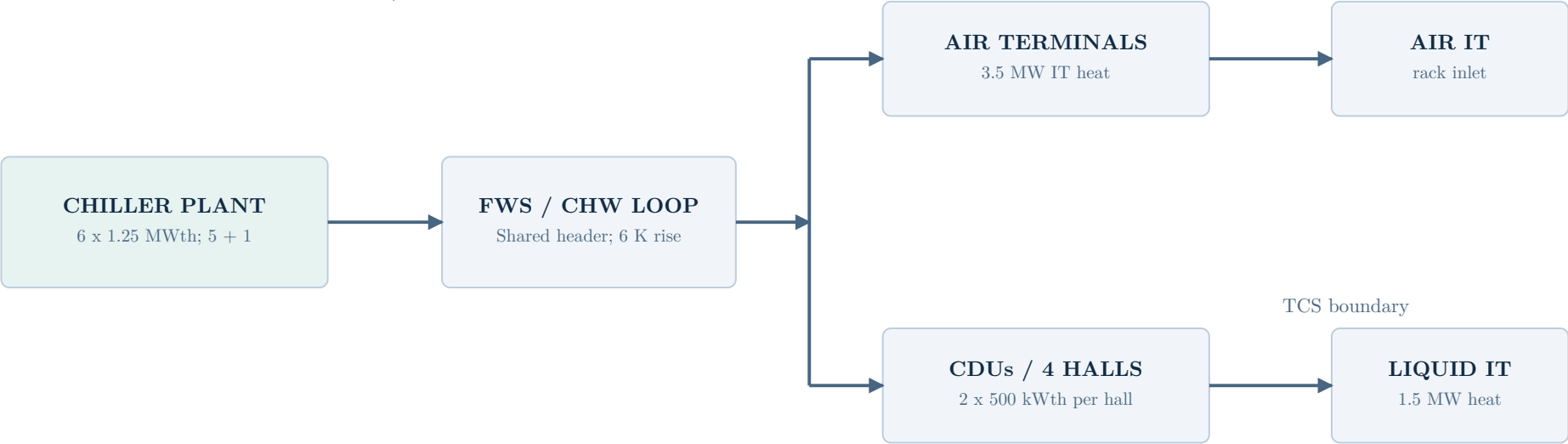
Each A/B path has a full-facility capacity allowance with one local unit unavailable. Total installed A+B nameplate is redundancy, not twice the customer demand. Supporting loads connect outside the IT UPS boundary; their detailed feeders are omitted here.

Battery allowance: **1,370 kWh per path** for ten minutes at modeled UPS output, after inverter, usable-energy and end-of-life factors. This is not a battery-string selection or cooling ride-through result.

Outstanding: utility/fault data, breaker/tie arrangement, protection coordination, earthing, cable sizing, generator dynamics, physical route separation and auxiliary power.

03 / Cooling and interface design

Facility cooling interfaces / conceptual supply direction



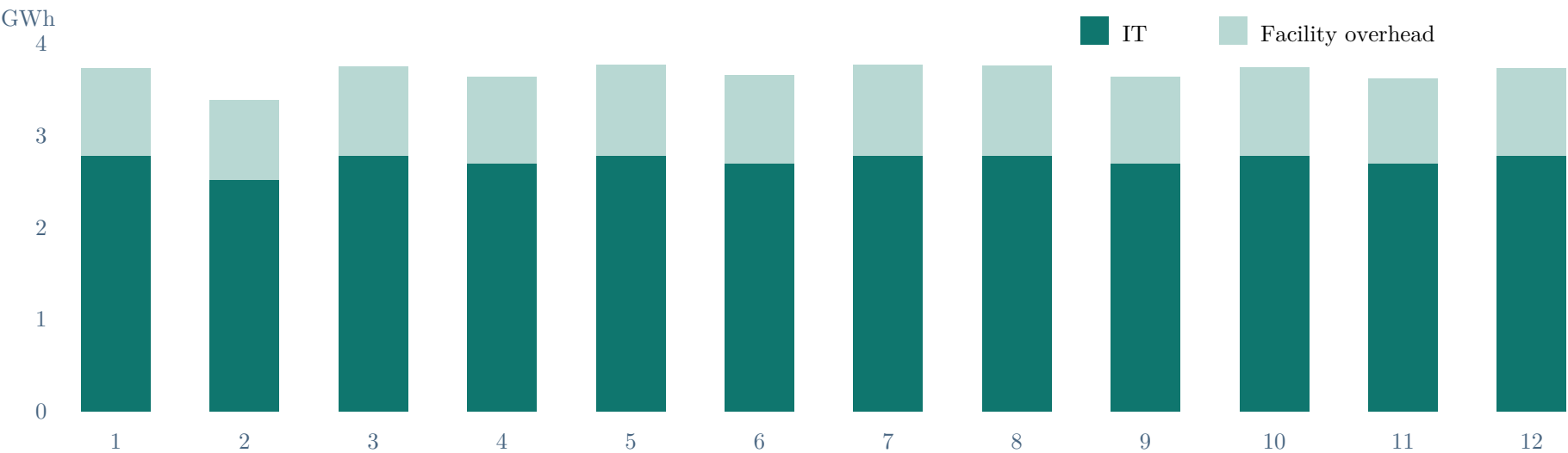
Concept flow diagram | supply/return collapsed; isolation, hydraulics, chemistry and thermal ride-through remain open.

QUANTITY	SCREENING RESULT	BOUNDARY
Total design cooling load	5,384 kWth	IT, UPS/distribution losses, terminal fans, envelope
CHW mass flow	214.4 kg/s	Water approximation; 6 K rise
Air-side volume flow	246.4 m3/s	Air IT + terminal fans; 12 K rise

The shared loop is an explicit common-path risk. Hydraulic isolation, water chemistry, condensation limits, pump selection and thermal transients need further design. The displayed layout is an original concept, not a copied vendor P&ID.

04 / Energy and sustainability

Annual gross facility energy includes IT plus supporting systems



Annual IT energy is 32.85 GWh and gross facility energy is 44.24 GWh. PUE is the ratio of annual energies, not an average of hourly ratios. Gross facility demand remains unchanged by optional PV accounting.

At the illustrative factors, grid emissions are 30,969 tCO2e/year and direct-site water is 1,642 m3/year. These are factor-based scenario outputs, not measured Indonesia performance. Baseline PV is zero; BESS has no annual dispatch model.

Excludes humidity/latent loads, vendor part-load maps, embodied materials, refrigerants, diesel and upstream water. Source definitions: DOE/FEMP guidance, S01 and S14 in the source register.

05 / Economics and sensitivity

NPV at the 10% nominal discount rate. Both drivers change the underlying cash-flow model; maintenance and residual value follow the changed base CAPEX.

CAPEX / CAPACITY PRICE	80%	100%	120%
80%	USD -2.62m	USD +11.96m	USD +26.53m
100%	USD -14.90m	USD -0.33m	USD +14.25m
120%	USD -27.19m	USD -12.61m	USD +1.96m

Power charges use imported energy and monthly demand peaks. In the 100% reimbursement base case, electricity costs cancel against recovery in net cash flow. This does not remove the gross utility expense or credit/contract risk.

Replacement allowance occurs in year eight; residual asset recovery occurs in year fifteen. The model has no debt, tax, working capital or construction-period cash flow. Test zero residual, lower reimbursement and delayed occupancy before interpreting financial attractiveness.

Method reference: NIST HB 135e2025 for lifecycle discounting concepts; all prices, rates and terms are synthetic project assumptions. The editable workbook exposes inputs, formulas, sensitivity and checks.

06 / Lifecycle coverage and honest maturity

LIFECYCLE	DELIVERED	NEXT EVIDENCE GATE
Define / select	OPR, assumptions, site gates, charter	Owner and real-site evidence
Value / design	Sizing, energy, cash flow, original concepts	Quotes, calibrated curves, detailed studies
Coordinate	DXF/SVG and rack-only OpenUSD with matching IDs	Survey, coordinated BIM and validated CFD
Control / assure	Points, six offline cases, FMEA and common-cause model	Hardware tests and topology validation
Build / commission	Procurement gates, FAT/SAT/IST plan	Witnessed physical evidence
Operate / improve	SOP/MOP/EOP, maintenance and dashboard specification	Site approval, drills and historian integration

The repository retains the requested lifecycle structure and adds a shared calculation package, strict schema, requirements traceability, source/license records, automated tests and hashed evidence. Each claimed maturity level has an explicit limit.

The portfolio emphasizes a reviewable chain: requirement -> input -> equation -> equipment/geometry -> risk -> verification. Folders that specify future CFD/BIM work are labeled as specifications, not completed solver results.

07 / Commissioning and operational evidence

CASE	OBSERVATION	EXPECTED SUPERVISORY STATE	RESULT
IST-001	Normal supply	UTILITY	PASS
IST-002	Utility lost	BATTERY	PASS
IST-003	Generator ready	GENERATOR	PASS
IST-004	No remaining supply	UNSERVED	PASS
IST-005	Telemetry stale	UNKNOWN	PASS
IST-006	Leak and high inlet temperature	UTILITY	PASS

These are six offline software checks, not physical integrated tests. They confirm state/alarms and the disabled-write boundary. Real acceptance requires approved procedures, safe prerequisites, instruments, witness records, deviations and operations sign-off.

Operational handover links asset records, current drawings, manuals, settings, spares, training, maintenance restrictions and incident procedures. A stale-data condition produces UNKNOWN, preventing a healthy-looking dashboard from being treated as trustworthy evidence.

The local model tests also cover dimensional balances, adverse inputs, PV curtailment, financial sign patterns, geometry IDs and common-cause availability. Remote CI remains pending user upload.

08 / Primary sources and access boundaries

S01 · Best Practices Guide for Energy-Efficient Data Center Design

DOE/FEMP; Otto Van Geet and David Sickinger, NREL; LBNL host | 2024-07

Public PDF read; broad design and metric guidance; no local compliance claim.

S02 · Data Center Tier Certification

Uptime Institute | undated

Public overview read; complete standards not reviewed.

S03 · Achieving High Reliability, Energy Efficiency in Data Center Design, Operations

ASHRAE | 2021-03-23

Public article read; full fifth-edition book not accessed.

S04 · AI Data Center Framework: Tools, Standards, and Resources

ASHRAE | undated

Directory read; encyclopedia substantive content inaccessible.

S05 · Data Center Liquid Distribution Guidance and Reference Designs, Revision 0

Open Compute Project | undated

Public PDF read; CC BY-SA 4.0; cite only, assets not redistributed.

S06 · Colocation Facility Guidelines for Deployment of Open Racks v4.0

Open Compute Project | 2020-12-15

Public PDF read; inconsistent BY-SA/BY notices; no copied assets. Covers Open Rack v1/v2.

S07 · Omniverse DSX Blueprint for AI Factories

NVIDIA | live repository

README reviewed; NVIDIA license terms; NGC assets separate; runtime not executed.

S08 · Data Center Design with Simscape

MathWorks | 2025-2026

BSD-3-Clause examples; MATLAB R2025b and listed commercial products required; not executed.

09 / Primary sources and access boundaries

S09 · Data Center Reference Design 100

Schneider Electric | 2026-03-14, version 3.0

Official landing metadata read; PDF download unavailable; no redistribution license established.

S10 · Data Center TradeOff Tools

Schneider Electric | undated

Catalog read; no calculator parity test or public source-code/API confirmed.

S11 · SustainDC: Benchmarking for Sustainable Data Center Control

Avishek Naug et al.; Hewlett Packard Labs | 2024; arXiv v5 2025-04-30

Paper record and PDF read; paper CC BY-NC-ND 4.0; no adapted figures.

S12 · SustainDC code

Hewlett Packard Enterprise | 2024; live repository

MIT code; separate scientific/ML dependency stack; not executed.

S13 · Life Cycle Costing Manual for FEMP, NIST HB 135e2025

Joshua Kneifel and David Webb; NIST | 2025-08

Public PDF inspected; supersedes 2022. US program parameters not used as Indonesian market inputs.

S14 · Cooling Water Efficiency Opportunities for Federal Data Centers

US Department of Energy | 2019-01-09

Public page read; historical efficiency examples not used as current benchmarks.

S15 · Omniverse Licensing

NVIDIA | 2026-09-02

Current licensing page reviewed; distribution remains governed by NVIDIA terms.

Accessed 3 September 2026. Research inspected relevant sections and source records, not every page of every cited work. Vendor runtimes and calculators were not executed. Full current ASHRAE content and the RD100 PDF remained inaccessible. Original project prose, diagrams and data are provided; external assets retain their own terms.